Scalability & Future Plan

Date	03.10.2026
Team ID	SWTID-2026-5957
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state using ServiceNow developer automation and advanced client-side scripting frameworks.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Client Scripts execute synchronously on form load, potentially affecting form rendering speed during heavy data traffic.	Slower incident form load times	High
2	UI Policies are scoped strictly to the Incident table without cross-table inheritance or dynamic multi-form mapping.	Limited reuse across task tables	Medium
3	Manual testing required for validating complex nested conditional UI policies and client script combinations.	Higher risk of script regression bugs	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports standard concurrent ITIL agents on single developer instance.	Implement asynchronous GlideAjax and optimized caching to handle thousands of concurrent requests smoothly.
2	Data Storage	Incident form fields tied to base ServiceNow relational database tables.	Adopt indexed queries and archive historical incident records to maintain high database retrieval performance.
3	Performance	Client-side validation scripts run on every field change event.	Refactor scripts to use debouncing and conditional execution filters to minimize browser resource usage.
4	Security	Role-based access control combined with basic client-side read-only rules.	Integrate advanced Access Control Lists (ACLs) with server-side script validation for robust enterprise data security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Core Client Script & UI Policy deployment for Incident form automation.	Q4 2026	Streamlines data entry and eliminates manual form errors.
Phase 2	Extend UI policies and client scripts to Problem and Change Request tables.	Q1 2027	Ensures consistent form behavior across all ITIL ITSM modules.
Phase 3	Incorporate automated unit testing frameworks for client-side scripts (ATF).	Q2 2027	Reduces regression testing time and ensures script reliability.
Phase 4	Integrate AI-driven predictive intelligence for automated incident field recommendations.	Q3 2027	Accelerates incident resolution and enhances user experience.